



Tesla Model S is a globally-popular EV

residential parking spaces as per requirement. Normal charging can be done across generic 16-ampere power source on walls. Charging cables need to be attached to the power source from one end and to the car from another. Car vendors can provide other appliances that may be necessary for charging, if needed. Some cars support fast charging, which may need an upgraded power source.

Some private players are bringing customised solutions for residential EV charging. For instance, talking about Magenta Power's ChargeGrid community charger, Lewis says, "Effectively, a housing society can deploy this charger at select parking slots and can recover the cost of power and solution. This allows for multi-user utilisation of the charger and charging space, thus drastically reducing the cost of deploying the charging solution."

Maintenance

EVs come at higher upfront costs but are easier to maintain in the long run for various reasons as compared to gas-driven cars. For starters, unlike gas-driven cars, EVs have no internal combustion engine as such. Car-owners will know that this means they can eliminate costs related to aging engines, like changing engine oil, transmission fluids, engine parts, coolants, etc, which become more frequent as engines age. This also keeps vehicle performance intact for a much longer period.

Comparing fuel costs, an EV can be massively beneficial, especially considering the average ₹ 80 per litre cost of petrol today. Considering

a basic assumption of a person driving 25 days a month, 50km every day, petrol can cost more than ₹ 6000 per month. The same amount of travel can be covered for just ₹ 1000 per month of electricity cost to charge an EV, assuming the average cost of electricity to be

₹ 7.5 per unit—these calculations vary in accordance to region, environment and extent of use.

Energy reuse technology of EVs is much better decorated as well, saving almost sixty per cent higher energy. Noiseless engines add to these advantages.

However, the main cost comes in the form of battery for EVs. The battery may have to be changed after six to eight years and can form nearly 35 per cent of the upfront cost, which can be a heavy bill.

Challenges with EVs

The main reason why EVs have not kicked off as well as we would have wanted them to is the upfront cost compared to performance. The top speed of EVs is lower and re-fuelling takes much longer for EVs. Lack of public charging infrastructure also raises the concern of charging on the road when battery runs out.

The market in India is also not as robust as it is in the rest of the world. Consumers are not yet prepared to pay the much higher cost for EVs. Importing a car costs a fortune, and even if one can afford to do so, the after-sales service is a major pain area.

However, the next few years, especially 2020, look to be quite promising, as many new players including Volvo, Tata, Kia, BMW, Audi and Hyundai have major plans to launch their EVs. The government is also working with private players to build widespread charging and battery purchase outlets. Once the market sets the tone, prices will slowly start to come down. **EFY**

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